CHAPTER

Motion-Corrected Reconstruction 13

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13.1 Introduction

Several methods are described in previous chapters that can be applied in the context of patient motion, and these are often referred to as dynamic imaging methods. Firstly, highly undersampled acquisition of the k space or k-t space can be used, in combination with sparse reconstruction methods, in order to make the acquisition time short with regard to patient motion (this is also referred to as imaging with high temporal resolution). In that case, "real-time imaging" is achieved, and the impact of motion may be neglected. However, a high temporal resolution is generally achieved at the cost of reduced spatial resolution. Secondly, temporally resolved imaging is a common alternative that consists of synchronizing the MRI data to certain physiological signals, such as the electrocardiogram or a respiratory motion signal. Such synchronization can be performed either at the acquisition stage, by prospectively selecting which k-space samples are acquired depending on which motion state we are in. It can also be performed at the reconstruction stage, by retrospectively sorting out the k-space data using extra motion information that was acquired simultaneously with the k-space data. The drawback is that such methods assume a high level of reproducibility of motion throughout the acquisition.

Motion correction methods have been further developed to enable k-space data acquired from multiple motion states to be combined. Similar to synchronization methods, motion correction in MRI can be prospective or retrospective. Prospective correction consists of modulating, in real time, the pulse sequence waveforms, i.e., the radiofrequency (RF) pulses and the magnetic field gradient pulses, so that the spatial encoding of the image accounts for the deformation of the imaged object or subject, thus prospectively "inverting" the effect of motion [65]. However, since the spatial encoding uses linear magnetic-field gradients, only linear changes of the coordinate system can be performed with such techniques, which correspond to affine transformations (i.e., a global translation, rotation, and scaling). Retrospective correction consists of "inverting" the effect of motion at the image reconstruction stage, and it is the focus of this chapter. One of the main advantages of retrospective correction over prospective methods is that it is not limited to affine motion, i.e., deformable motion can also be addressed, as we will see.

Whether we consider the acquisition of a static image or a dynamic scene, the reconstruction methods that we have seen so far implicitly assume that the imaged organs are static during the acquisition of the k-space data of each image to be reconstructed. To ensure this assumption holds in the clinical practice, the patient is asked to remain still during the sequence, for a period ranging from a few seconds to several minutes. Specifically, in cardiac or abdominal MRI protocols, patients are asked to hold their

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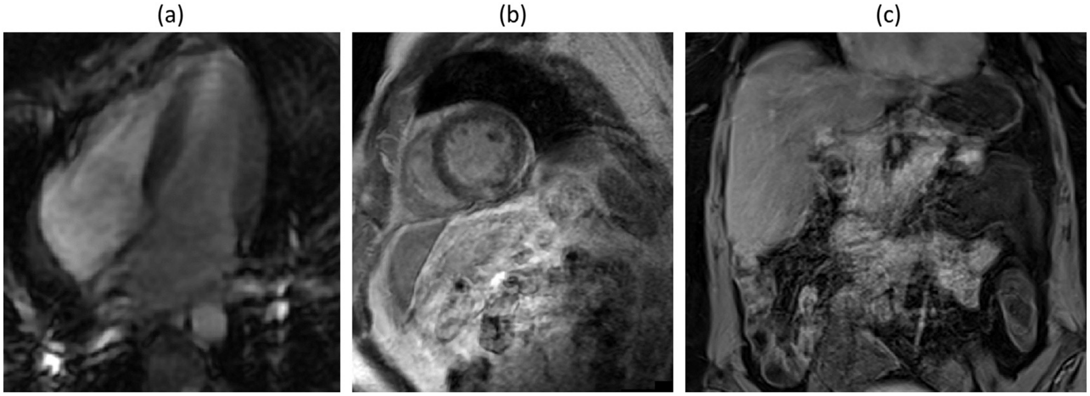

FIGURE 13.1

Example motion artifacts in patients: cardiac cine MRI (balanced SSFP sequence, systolic phase) from a eight-year-old Duchenne muscular dystrophy patient unable to hold his breath (**a**), showing blurring throughout the image and ghosts near the ventricular apex; 3D late gadolinium enhancement sequence (inversion-recovery fast spoiled gradient echo) from an adult cardiac patient with poor breath-holding (**b**), showing blurred myocardium hindering the precise delineation of the myocardial scar; abdominal slice from an Crohn disease patient referred for MR enterography (3D late gadolinium enhanced VIBE sequence) obtained after poor breath-holding (**c**), showing blurring/ghosting of all abdominal structures, including liver and bowel.

breath repeatedly for periods of 10 to 20 s. Some patients, including elderly, pediatric, or psychiatric patients, are less likely to comply with those instructions. Moreover, there are some types of motion that simply cannot be avoided (e.g., heart beating). Motion can occur at random times (e.g., involuntary head motion, swallowing, coughing) or it can be pseudo-periodic (heart beating, breathing, peristalsis). The most obvious manifestation of motion is a degradation of the image quality (see Fig. 13.1). As we know, the MRI data is sampled in the Fourier domain, thus the resulting artifacts in the image domain due to motion in k-space are quite complex, in that they are not only localized in the area of the moving structure. This can alter the quality and diagnostic value of the image. As a result, it can also lengthen the examination due to the need to repeat corrupted scans, and ultimately degrade the patient experience with MRI. Besides this direct impact, motion also has an indirect impact on clinical MRI. Indeed, since most patients cannot hold their breath for much longer than 20 s, cardiac and abdominal imaging protocols are designed with a compromised image quality, typically using a much lower spatial resolution (especially in the slice direction) than in static organs. Thus, reconstruction methods from free-breathing data can therefore enable new applications as shown by several approaches that we will discuss in this chapter.

In this chapter, we will focus on estimating and integrating motion into the forward acquisition model E, introduced in Chapter 2. To achieve that goal, we will need to make a number of assumptions pertaining to MRI physics. First, we will assume that motion between the RF excitation and the sampling of k-space data—also termed intraview motion—can be neglected.[1] This assumption is often reasonable in practice when motion-induced changes are slow compared to the echo time of the imaging sequence. As a consequence, motion can be considered to occur between successive phase encoding steps—this is also known as interview motion—and only the spatial encoding process is affected. Due to inconsistencies in the imaged content during the acquisition of the different sets of k-space samples, a conventional reconstruction by inverse Fourier transform results in a combination of blurring (in all directions of motion) and ghosting artifacts (in the phase-encoding direction(s)) [92]. Such artifacts are illustrated in Figs. 13.1 and 13.2, and are precisely what we seek to suppress, or at least minimize, with the motion corrected reconstruction techniques described in this chapter. Furthermore, it will be assumed that a motion-corrected image does exist, which means that there exists a reference motion state from which any motion state can be derived by a spatial deformation of the imaged organs/tissues, and only by a spatial deformation. In terms of physics, this assumption can be thought of as a signal conservation rule—the signal being what is "seen" by the imaging system, analogous to mass conservation. Therefore, large through-plane or out-of-volume motion, contrast changes during the course of the sequence, or changes of the magnetization due to spin history effects should be treated carefully. This assumption should also be viewed from a physiological standpoint. For instance, the reconstruction of phase-contrast cardiac MRI data acquired during free breathing uses data from multiple respiratory motion states that may correspond to different velocities.

With that in mind, we will first consider the problem of MRI reconstruction in the presence of known motion. Here "known motion" means that we assume motion has already been estimated or guessed by any means. This can be, for instance, from real-time motion measurements of bulk head motion, from a low-resolution image navigator, or from a predictive model combining real-time motion signals and prior imaging data. Because this motion estimation step is generally not perfect, we will next consider motion as an additional optimization variable for the reconstruction problem. Provided motion can be modeled and parameterized, the reconstruction can then be reformulated as a joint estimation of a static image and some motion parameters. In the methods section we will mainly review the strategies that can be used in practice to provide estimates of motion. Then a few applications will be described, with the objective to illustrate how these methods can be fit to various settings, i.e. using either extra hardware for motion sensing, or MR navigator data, or nothing but the raw k-space data. Finally, we will discuss some of the remaining challenges, and, in particular, we will come back to the validity of the various assumptions and possible directions for future research.

[1] This assumption is invalidated, for instance, when fast motion occurs during a conventional fast spin echo or a diffusion-weighted sequence. Certain types of motion (e.g., pulsatile motion in the brain, cardiac contraction) may cause intravoxel dephasing, leading to signal dropouts. In some cases, a sequence design using gradient moment-nulling techniques can minimize such effects.

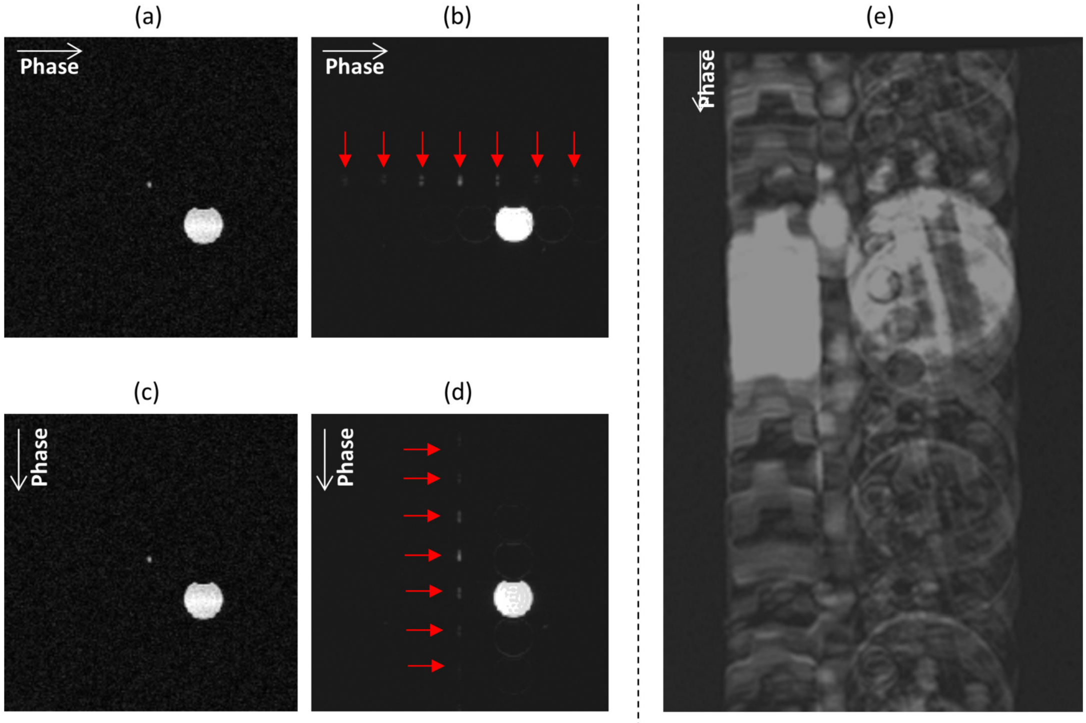

FIGURE 13.2

Example motion artifacts in phantoms subjected to a controlled, periodic, translational motion in the vertical direction. The left-hand figures show spin-echo images without (respectively with) motion of a small gadolinium sphere (at the center of the image), when the phase encoding direction is horizontal (**a** and **b** respectively) or vertical (**c** and **d**). Artifacts appear as replicas (ghosts) in phase direction and localized blurring in the direction of motion. The right-hand figure shows another moving phantom, during a fast spin-echo sequence with interleaved k-space sampling.

13.2 Theory

13.2.1 Reconstruction with known motion: the particular case of translational motion

In the particular case of a bulk translational motion of the imaged object/subject, the motion correction problem may be simplified using properties of the Fourier transform. Let $\boldsymbol{\tau}_{\boldsymbol{k}} = \begin{bmatrix} \tau_{k,x} & \tau_{k,y} & \tau_{k,z} \end{bmatrix}$ be a translation vector affecting the image during the acquisition of a given k-space sample of coordinate $\boldsymbol{k} = \begin{bmatrix} k_x & k_y & k_z \end{bmatrix}$. The MRI signal for that k-space sample, for a given receiving coil c (assumed to be static), is the Fourier transform of the shifted object, weighted by the coil sensitivity (defined in

Chapters 2 and 6 for single and multiple coils, respectively)

$$s_c(\boldsymbol{k}) = \iiint C_c(\boldsymbol{r})\, x(\boldsymbol{r} - \boldsymbol{\tau}_{\boldsymbol{k}})\, e^{-2i\pi \boldsymbol{k}\cdot\boldsymbol{r}} \boldsymbol{dr}. \tag{13.1}$$

Using the substitution rule $\boldsymbol{r}' = \boldsymbol{r} - \boldsymbol{\tau}_{\boldsymbol{k}}$ in the integral,

$$s_c(\boldsymbol{k}) = e^{-2i\pi \boldsymbol{k}\cdot\boldsymbol{\tau}_k} \iiint C_c\left(\boldsymbol{r}' + \boldsymbol{\tau}_{\boldsymbol{k}}\right) x\left(\boldsymbol{r}'\right) e^{-2i\pi \boldsymbol{k}\cdot\boldsymbol{r}'} \boldsymbol{dr}'. \tag{13.2}$$

If we further assume that spatial variations in the coil sensitivity are small for the given motion, we can make the approximation that $C_j\left(\boldsymbol{r}' + \boldsymbol{\tau}_{\boldsymbol{k}}\right) \approx C_j\left(\boldsymbol{r}'\right)$, leading to a simplified expression

$$s_c(\boldsymbol{k}) \approx e^{-2i\pi \boldsymbol{k}\cdot\boldsymbol{\tau}_k} F(C_c x). \tag{13.3}$$

It follows that each coil image $C_c x$, rewritten as x_c, can then be reconstructed directly using the inverse Fourier transform of the k-space data, multiplied by phase-correction terms for each k-space sample

$$x_c(\boldsymbol{r}) \approx \iiint s_c(\boldsymbol{k})\, e^{2i\pi \boldsymbol{k}\cdot\boldsymbol{\tau}_k} e^{2i\pi \boldsymbol{r}\cdot\boldsymbol{k}} \boldsymbol{dk}. \tag{13.4}$$

Finally, the image is reconstructed by a coil combination technique. Conventional methods include the sum-of-squares reconstruction, or the SENSE reconstruction with an acceleration factor of 1 [78], i.e., $x = \sum_{c=1}^{N_c} C_c^* x_c / \sum_{c=1}^{N_c} |C_c|^2$.

This approach can be extended to rotational motion or affine motion [73]. Affine motion can be defined by a transformation of the form $\boldsymbol{r} \mapsto \mathcal{A}\boldsymbol{r} + \boldsymbol{\tau}$, with $\mathcal{A}$ a 3×3 matrix (which can describe rotation, shearing, scaling), and the translation vector $\boldsymbol{\tau} = \begin{bmatrix} \tau_x & \tau_y & \tau_z \end{bmatrix}$. In particular, a rotation in image space is equivalent to a rotation by the same angle in k-space. A regridding may be necessary in that case, as well as a density compensation (see Chapter 4). However the general formulation described in the next section will be preferred, even in the case of simple motion (translations, rotations), for several reasons: (i) This k-space correction method makes an additional assumption about coil sensitivities as we have seen, which may lead to errors; (ii) the proposed reconstruction formula in Eq. (13.4) is an empirical method rather than a mathematically well-defined inversion procedure; and as a result it does not provide an optimal solution to the motion correction problem, e.g., in the least-squares sense. We will come back to this point in the next section, and the differences between the two approaches are illustrated in the practical tutorial provided with this chapter.

13.2.2 Reconstruction with known motion: the general case

In this section we seek to integrate "arbitrary" patient motion in the forward acquisition model E of the linear inverse problem formulation of MRI reconstruction. By "arbitrary", we mean that it can be a simple translation or a rigid transformation of the image, but it can also consist of free local deformations, which will be termed elastic or nonrigid. Physically though, motion is not completely arbitrary. The displacement fields describing subject motion in the field-of-view can be assumed to be continuous and smoothly varying in space, except potentially at the interface between certain organs exhibiting a sliding motion (e.g., at the interface between the liver and abdominal wall), or at the interface between cavity and blood for instance. Importantly, we assume that the imaged organs remain mostly within the imaged slice or volume during the acquisition, which means that, for 2D imaging, only in-plane motion will be considered, and through-plane motion will be disregarded. In the particular case of 2D imaging, through-plane motion may also result in an *apparent* in-plane displacement of the structures that is different for the true 3D motion, e.g., in a short axis slice the heart ventricles may appear to be a little bit larger or smaller due to the component of the 3D heart motion that lies in the longitudinal direction of the heart. Therefore, an apparent 2D displacement field may be used to model motion in that short-axis slice, which may actually combine structures coming from closely adjacent regions of the heart in the through-plane direction. Since relatively thick slices are generally used in cardiac imaging (8 mm, typically) the framework described here can also be useful in practice, but it should be kept in mind that what is corrected for is this apparent 2D motion.

13.2.2.1 Motion operators

In the mechanics of continuous media, motion is described by a displacement vector field $\boldsymbol{u}(\boldsymbol{r},t)$, which means that, for each point $\boldsymbol{r}$ in space, its displacement vector at time t is $\left[u_x(\boldsymbol{r},t) \quad u_y(\boldsymbol{r},t) \quad u_z(\boldsymbol{r},t)\right]$. The intensity of the MR image at a given time t is therefore given by $x(\varphi(\boldsymbol{r},t))$, with the mathematical transformation $\varphi : \boldsymbol{r} \mapsto \varphi(\boldsymbol{r},t) = \boldsymbol{r} + \boldsymbol{u}(\boldsymbol{r},t)$. The question that follows naturally is: How can we move on from this mechanics formulation of the spatial transformation φ to a matrix formalism that is better suited for image reconstruction problems as we have seen in previous chapters, and which would allow us to apply some numerical solvers? We want to formulate the operator $M_u(x)$ that takes the image matrix x as an input, applies the displacement field u to it, and returns the transformed image matrix. If we try, "naively", to apply u to the spatial coordinates of each voxel of a digital image x, we see that M_u has to affect the intensity value at the reference motion state, $x(\boldsymbol{r})$, to a voxel near the coordinate $\boldsymbol{r} + \boldsymbol{u}(\boldsymbol{r},t)$. This approach has a major drawback: For certain types of displacement fields, e.g., a local expansion, it will leave some "holes" in the transformed image matrix, i.e., there is no guarantee that each voxel of the transformed image will be filled in with a proper value.

To solve this problem, a more convenient way is generally used in the field of digital image processing, in particular for image registration, and that is illustrated in Fig. 13.3. The transformation is described by the inverse displacement field instead, i.e., we will now consider that u is the displacement field from the target motion state to the reference motion state [68]. By doing so, the transformed image $M_{\boldsymbol{u}}(x)$ can be formed by searching, for each of its voxels r, the corresponding intensity value which is $x(\boldsymbol{r} + \boldsymbol{u}(\boldsymbol{r},t))$. The intensity values in image x are only known at the nodes of a regular grid so, in general, $\boldsymbol{r} + \boldsymbol{u}(\boldsymbol{r},t)$ does not lie exactly on one of these particular nodes. Therefore, $x(\boldsymbol{r} + \boldsymbol{u}(\boldsymbol{r},t))$ is estimated by interpolation, which means that it is estimated by a weighted sum of the intensity values in x at the closest nodes. As a result, the operator $M_{\boldsymbol{u}}$ is a linear function of x and can be described by a matrix. Therefore, in the remainder, we will note the transformed image as the matrix–vector product $\boldsymbol{M_u x}$. Each row of the matrix $\boldsymbol{M_u}$ is responsible for calculating the intensity value of one particular voxel of the transformed image. In the case a 2D linear interpolation kernel is used, the closest voxels form a 2×2 square so this row of $\boldsymbol{M_u}$ will comprise four nonzeros elements, which are the four interpolation weights. These four elements will be located in the columns of $\boldsymbol{M_u}$ corresponding to the indices of the four surrounding voxels in x (see Fig. 13.3).

Using this explicit matrix formulation of the motion operators, its transpose can be computed exactly, without any approximation (the transpose operators are also required for solving the inverse problem). Indeed, some authors have used the approximation $\boldsymbol{M_u^T} \approx \boldsymbol{M_{-u}}$, but it is not always valid.

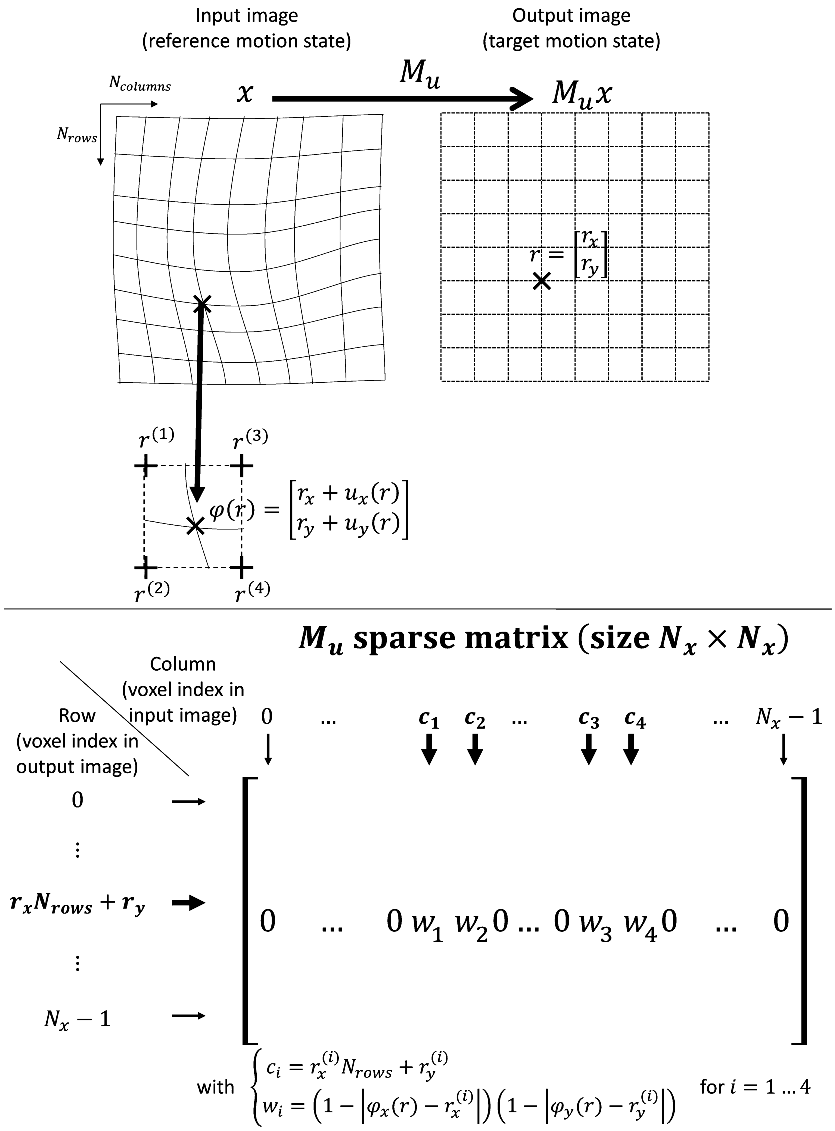

FIGURE 13.3

Definition and implementation of the motion operator $\boldsymbol{M_u}$ used to transform an image from a reference to a target motion state, knowing the underlying displacement field $\boldsymbol{u}$.

13.2.2.2 Forward acquisition model including motion operators

Once the motion operators have been determined, forming the forward acquisition model is relatively straightforward.

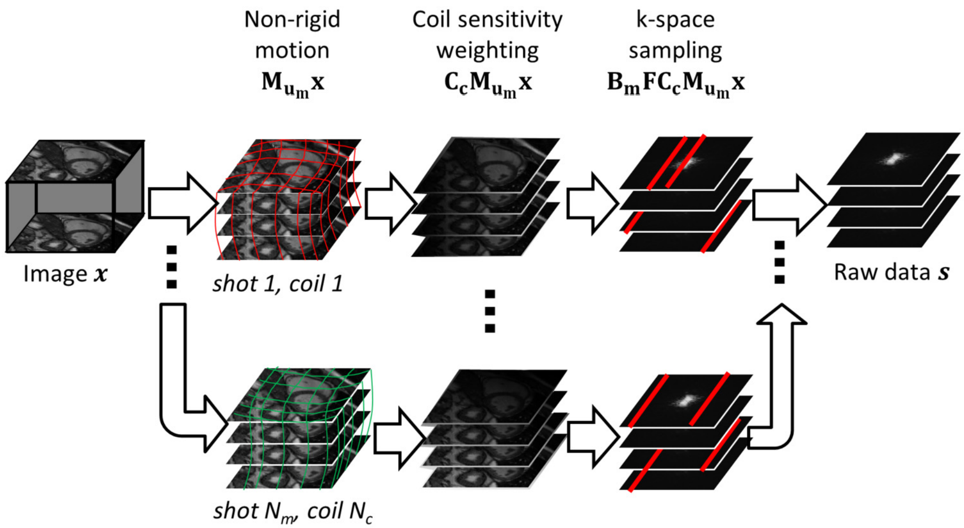

FIGURE 13.4

Forward model of MRI acquisition in the presence of known motion between successive phase-encoding steps or shots, including the motion operators.

Let us assume that the MR acquisition comprises N_m shots (i.e., N_m subsets of the full k-space scan) and that motion occurs between each of these shots, as described by a motion operator $\boldsymbol{M}_{\boldsymbol{u}_m}$ ($m = 1 \ldots N_m$) that takes the imaged subject from a reference motion state to its motion state at shot m. The reference motion state can be chosen arbitrarily, but it is often convenient to define it as an "average" position, or possibly a position where data are more frequently acquired (e.g. the end-expiratory plateau of a free-breathing scan). As we have just discussed, the underlying displacement field $\boldsymbol{u}_m$ defines the mapping of physical coordinates from the m^{th} motion state to the reference motion state.

In the most general case, each shot contains only one k-space readout (e.g., one line in a Cartesian acquisition) and N_m is the number of k-space lines. A shot can also contain several lines of k-space acquired within a very short time period, during which motion is assumed to be negligible. In a more computationally efficient implementation, a shot can be a grouping of k-space lines acquired in a similar motion state (e.g., same phase as the respiratory cycle). In the latter case, N_m is the number of distinct motion states and motion within each shot is disregarded.

Given these definitions, the forward acquisition model is constructed in a way to simulate the motion-corrupted scan, given a static image $\boldsymbol{x}$ (in the reference motion state), and to return the motion-corrupted k-space data $\boldsymbol{s}$

$$\mathbf{s} = \boldsymbol{E}\boldsymbol{x}. \tag{13.5}$$

Therefore, the encoding operator $\boldsymbol{E}$ is, as illustrated in Fig. 13.4, the concatenation of k-space samples acquired by each shot or motion state (with $\mathbf{B}_m$ the sampling matrix at the motion state/shot m) and by each coil receiver, which are the Fourier transforms ($\boldsymbol{F}$ being the Fourier transform operator) of the deformed image $\mathbf{M}_{\mathrm{u}_n}\boldsymbol{x}$, weighted by the sensitivity of the coil receiver $\boldsymbol{C}_c$ [9,68]

$$\mathbf{E} = \left[\begin{array}{c} \mathbf{B}_1\mathbf{F}\mathbf{C}_1\mathbf{M}_{\mathrm{u}_1} \\ \vdots \\ \mathbf{B}_{\mathrm{N}_m}\mathbf{F}\mathbf{C}_1\mathbf{M}_{\mathrm{u}_{\mathrm{N}_m}} \\ \hline \vdots \\ \hline \mathbf{B}_1\mathbf{F}\mathbf{C}_{\mathrm{N}_\mathrm{c}}\mathbf{M}_{\mathrm{u}_1} \\ \vdots \\ \mathbf{B}_{\mathrm{N}_m}\mathbf{F}\mathbf{C}_{\mathrm{N}_\mathrm{c}}\mathbf{M}_{\mathrm{u}_{\mathrm{N}_m}} \end{array}\right]. \tag{13.6}$$

Let us explicitly give the size of the matrices composing $\boldsymbol{E}$. The motion operators $\mathbf{M}_{\mathrm{u}_m}$ $(m = 1 \ldots N_m)$ are sparse matrices as described in the previous section, of size $N_x \times N_x$, where N_x is the number of voxels is the image. The coil sensitivity operators $\mathbf{C}_\mathrm{c}$ $(c = 1 \ldots N_c)$ are diagonal matrices of size $N_x \times N_x$, and the diagonal elements are the complex sensitivity values of coil c. The Fourier transform operator $\boldsymbol{F}$ (which can be either 2D or 3D) is a dense matrix of size $N_x \times N_x$. In a Cartesian acquisition, it is the classic discrete Fourier transform, but in a non-Cartesian acquisition it can be defined as the nonuniform fast Fourier transform (NuFFT) operator. The sampling operators $\mathbf{B}_\mathrm{m}$ $(m = 1 \ldots N_m)$ are sparse matrices with values 0 or 1. Each $\mathbf{B}_\mathrm{m}$ matrix has N_x columns, and its number of rows is the number of k-space samples acquired at shot m. Finally, the size of $\boldsymbol{E}$ is $N_c N_s \times N_x$, with N_s the total number of acquired k-space samples for each coil.

13.2.2.3 *Solving the inverse problem*

A motion-corrected image $\hat{\boldsymbol{x}}$ can be obtained by solving the inverse problem, using a classic least-squares optimization scheme

$$\hat{\boldsymbol{x}} = \arg\min_{\boldsymbol{x}} \|\boldsymbol{E}\boldsymbol{x} - \boldsymbol{s}\|_2^2 + \lambda \|\boldsymbol{x}\|_2^2 . \tag{13.7}$$

Here, for simplicity, a Tikhonov regularizer is used, but the framework can be easily adapted to include more advanced regularizers [83], such as those used to enforce sparse models (L_1 norms, low-rank models, etc.) which have been more thoroughly described in the previous chapters. As shown in Chapter 2, Problem (13.7) has the following solution:

$$\hat{\boldsymbol{x}} = \left(\boldsymbol{E}^H\boldsymbol{E} + \lambda\boldsymbol{I}\right)^{-1}\boldsymbol{E}^H s. \tag{13.8}$$

The inversion can be performed using a matrix-free solver, i.e. it does not require the full $\boldsymbol{E}$ matrix to be formed and stored explicitly, such as the conjugate-gradient algorithm. Algorithms to solve a problem of this type are discussed in Chapter 3.

The solution provided by Eq. (13.8) is a mathematical inversion of a large composite operator, which can be seen as the sum of several individual operators corresponding to the various motion states or shots in the acquisition. It is interesting to compare this approach to the k-space phase correction given in Section 13.2.1 in the case of a translational motion. The phase-correction method consists of applying an inverse motion operator for each shot individually (since it is equivalent to applying the inverse translation in image space for each shot), and then gathering the data from all shots. Therefore, it is the sum of motion-inverted k-space data rather than the invert of the sum of motion-corrupted k-space data. In general, the sum of inverses is different from the inverse of the sum, as observed in [9], and therefore the phase-correction method provides a suboptimal solution in the least squares sense.

In the particular case where no motion occurs, the motion operators are all equal to the identity matrix, and the reconstruction in Eq. (13.8) is a classic regularized SENSE reconstruction, as seen in Chapter 6. Similar to parallel imaging reconstruction problems, noise-correlation matrices can be incorporated to provide an SNR-optimal solution [79]. This can be done in a pre-processing step, prior to image reconstruction, by forming virtual receiver channels—with their associated virtual coil sensitivities and k-spaces—which have ideal noise decorrelation across channels [76]. To keep notations simple, in this chapter we will therefore assume receiver channels are ideally decorrelated.

Conditioning of the system

The condition number of the $\boldsymbol{E}^H\boldsymbol{E}$ matrix is larger than that a classic inverse Fourier transform or SENSE reconstruction because the introduction of the motion operators makes the linear system of equations deviate from an ideal set of linearly independent equations. The actual value of the condition number depends on the type of motion and on the k-space sampling pattern. The effect of these elements on conditioning can be illustrated by a simple thought experiment, as described in [7]. Let us consider an acquisition made of two shots or motion states: The first half of k-space is sampled during the first shot, the second half during the second shot, and there is a 90° rotation of the imaged object between the two shots. Because the 90° rotation in image space is equivalent to a 90° rotation in k-space, this means that we have effectively collected samples corresponding to three out of four tiles of the Fourier plane, and the fourth tile is left unsampled. This means there is not enough information to reconstruct the image exactly according to the Nyquist limit.

Apart from using regularization, one should also consider another convenient way to improve the conditioning of the system. If we acquire the k-space with several repetitions (i.e., with a number of excitations Nex > 1), provided that the same k-space samples are collected several times but in different motion states, they will provide new linearly independent data to the linear system of equations [68]. Unlike classic reconstructions, the motion-corrected reconstruction does not average the k-space data in the dimension of Nex, but instead it considers multiple Nex data as separate shots or motion states.

As a conclusion, in the ideal case of known motion, using a combination of repeated sampling (Nex > 1) and regularization, Eq. (13.8) provides a mathematically rigorous framework for motion correction and can provide very accurate solutions. The main difficulty that remains for its practical implementation is to accurately estimate motion. Several approaches will be described in Section 13.3 but they generally provide imperfect estimates. Therefore, one will also consider the case where motion is treated as an additional unknown of the reconstruction problem.

13.2.3 Joint reconstruction of image and motion

As discussed in Chapter 2 for nonlinear reconstructions in MRI, in this section we rewrite the motion-corrupted acquisition model $\boldsymbol{E}$ as $\boldsymbol{E}(\boldsymbol{u})$ in order to show explicitly its dependency on the displacement field $\boldsymbol{u}$. For the moment we make no assumption on $\boldsymbol{u}$, which can vary in space and time (the time dimension corresponding to the shots or motion states, as defined in the previous section). We now consider the following optimization problem to solve for the motion-corrected image and the motion

Table 13.1 Autofocus approach versus joint estimation of image and motion.

	"Autofocus" reconstruction	**Joint optimization of image and motion**
Main cost function to be minimized	$\mathcal{R}\left(\boldsymbol{E}\left(\boldsymbol{u}\right)^{-1}\boldsymbol{s}\right)$ e.g., $\mathcal{R}(x)$ = image entropy	$\\|\boldsymbol{E}\left(\boldsymbol{u}\right)\boldsymbol{x}-\boldsymbol{s}\\|_2^2$
Variables to be optimized explicitly	Motion parameters: u	- Motion parameters: u - Image: x
Variables to be optimized implicitly	Solution image $\hat{x}=E\left(\hat{u}\right)^{-1}s$ with $\hat{u}$ the solution of the motion parameter optimization problem	None
Optional regularizers to be added to the cost function	On motion parameters: $+\boldsymbol{\mu}\mathcal{S}(\mathrm{u})$ e.g., $\mathcal{S}(\mathrm{u})$ = spatial/temporal smoothing of motion parameters	- On the solution image: $+\boldsymbol{\lambda}\mathcal{R}\left(\boldsymbol{x}\right)$ e.g. $\mathcal{R}(x)$ = L2 norm, L1 norm, image entropy... - On motion parameters: $+\boldsymbol{\mu}\mathcal{S}(\mathrm{u})$
Approximations for practical solving	$E\left(u\right)^{-1}s\approx$ sum of motion-inverted k-space data (e.g., phase correction method described in Section 13.2.1.)	None

field:

$$\left(\hat{\boldsymbol{x}},\hat{\boldsymbol{u}}\right)=\underset{\boldsymbol{x},\boldsymbol{u}}{\arg\min}\,\|\boldsymbol{E}\left(\boldsymbol{u}\right)\boldsymbol{x}-\boldsymbol{s}\|_2^2+\lambda\,\|\boldsymbol{x}\|_2^2+\mu\,\|\boldsymbol{S}\boldsymbol{u}\|_2^2\,. \tag{13.9}$$

Here, we have introduced a second regularization term in order to impose a constraint on the displacement field $\boldsymbol{u}$. This term includes an operator $\boldsymbol{S}$, which will be specified later according to the additional assumptions that will be made on motion. Indeed, in the present form, the number of unknowns is extremely large. The image $\boldsymbol{x}$ has N_x elements, the displacement field $\boldsymbol{u}$ has $N_{dims}N_mN_x$ elements, with N_m the number motion states (or shots), and N_{dims} the number of dimensions of the image (2D or 3D). The number of collected k-space samples is N_cN_s. For example, if we consider a 256 × 256 image, acquired with 8 shots, 1 Nex, 32 coils, we have about 1.1 million unknowns and 2.1 million acquired data samples. However, there is a lot of redundancy within the 32 coils so, although the system appears overdetermined, it is likely to be severely ill-conditioned.

Relation to "autofocus" method

It should be noted that another formulation has been proposed in the literature, called autofocus that consists of optimizing motion parameters that minimize an image-quality metric, such as the image entropy [5,6,15,54,56]. Table 13.1 summarizes the main differences between the "autofocus" approach and the joint optimization approach. Autofocus methods also result in a joint optimization problem for the image and motion parameters, but the image reconstruction step is hidden, or embedded, in the motion estimation step (e.g., optimizing the image entropy of the motion-corrected image). Mathematically, optimizing the reconstructed image quality with respect to the motion parameters is a more difficult problem because the associated forward model (relating the motion parameters to the cost function) requires a model for the motion-corrected image reconstruction. However, this involves an inversion procedure, i.e., $\boldsymbol{E}(\boldsymbol{u})^{-1}s$, so a direct model of the reconstruction is not easily available. In practice authors used an estimate of the "exact" reconstruction, e.g., a phase-corrected k-space reconstruction in case of translational motion, as described in Section 13.2.1. This is conceptually close to assuming that $\boldsymbol{E}(\boldsymbol{u})^{-1} \approx \boldsymbol{E}(\boldsymbol{u})^T$. Though this approach can be effective in the rigid case, it does involve an approximation, and it is difficult (or computationally inefficient) to generalize it to nonrigid motion. Therefore, Eq. (13.9) will be preferred. Note that the image-entropy metric $H(x) = \sum_{n=1}^{N_x} |x_n|/x_{max} \log(|x_n|/x_{max})$ [5], with x_{max} the maximal intensity in the image, could be used in Eq. (13.9) instead of the classic regularizer $\|\boldsymbol{x}\|_2^2$. Actually, if we ignore the normalization by x_{max}, the image entropy can be thought of as an "intermediate" between the more familiar L_1 and L_2 norms, respectively $\|\boldsymbol{x}\|_1 = \sum_{n=1}^{N_x} |x_n|$ and $\|\boldsymbol{x}\|_2^2 = \sum_{n=1}^{N_x} |x_n|^2$.

Parameterizing motion

Instead of directly solving the general problem in Eq. (13.9), in this chapter we will seek to parameterize motion in order to reduce the number of unknowns significantly. If we choose a particular model for motion, the displacement field $\boldsymbol{u}$ can be described by a reduced number of parameters $\boldsymbol{\alpha}$, and therefore only those parameters need to be optimized. We will consider two motion models: The first one will be a spatially uniform displacement field, i.e., only global translations of the object/subject will be considered (reducing the number of motion parameters to $N_{dims} N_m$); the second one will allow spatially varying (i.e., nonrigid) displacements, under a temporal constraint that time variations of $\boldsymbol{u}$ are linear functions of a number N_k of known motion signals (reducing the number of motion parameters to $N_{dims} N_x N_k$ with $N_k \ll N_m$). Therefore, the joint reconstruction is reformulated as a joint optimization of the static image and the motion model parameters $\boldsymbol{\alpha}$

$$\left(\hat{\boldsymbol{x}}, \hat{\boldsymbol{\alpha}}\right) = \underset{x,\alpha}{\arg\min} \left\| \boldsymbol{E}\left(\boldsymbol{u}\left(\boldsymbol{\alpha}\right)\right) \boldsymbol{x} - \boldsymbol{s} \right\|_2^2 + \lambda \left\| \boldsymbol{x} \right\|_2^2 + \mu \left\| \boldsymbol{S} \boldsymbol{\alpha} \right\|_2^2 . \tag{13.10}$$

In case of a translational motion model, the regularization on $\boldsymbol{\alpha}$ may be omitted since the number of added parameters remains small. Alternatively, $\boldsymbol{S}$ can be chosen to be the identity operator or a temporal smoothing if there are good reasons to assume motion cannot vary too abruptly between two successive shots or motion states. In case of a temporally constrained nonrigid motion model, $\boldsymbol{S}$ can be the spatial gradient operator to ensure $\boldsymbol{\alpha}$, and thereby $\boldsymbol{u}$ in this case is spatially smooth.

Alternating optimization scheme

In order to solve the problem in Eq. (13.10), an alternating optimization scheme can be used. The main difficulty is that the cost function is a nonlinear function of $\boldsymbol{u}$ (and thereby of $\boldsymbol{\alpha}$). This problem belongs to the class of nonlinear least squares problems. Methods for solving this type of problems have been described in Chapter 3 and, specifically for motion-compensated reconstruction, have been described in [69,71] in the case of a nonrigid motion model and in [21,38] in the case of rigid motion. To keep the formalism consistent for motion-compensated reconstruction in the following sections, we will describe the Gauss–Newton strategy initially proposed for nonrigid motion models, and briefly discussed in Chapter 3, but we will also adapt it to the rigid case (translations only). The Gauss–Newton method consists of linearizing the cost function around the current estimate of motion. This turns the nonlinear least-squares problem into a sequence of linear least-squares problems, which we are more familiar with. This linearization step is described mathematically by the so-called Jacobian matrix. This matrix may seem a bit abstract so it may be useful to give a physical interpretation for it. It actually relates to how a small error in the motion model will propagate into the forward model that we have described in the previous section. In other words, it tells us how much our motion-corrupted acquisition model deviates from the actual k-space data measurements when a small perturbation of the motion model is applied.

13.2.3.1 Propagation of motion errors

Let us assume an estimate of motion $\boldsymbol{u}$ is available that is related to the true displacement field $\boldsymbol{u}_{true}$ as: $\boldsymbol{u}_{true} = \boldsymbol{u} + \delta \boldsymbol{u}$. We define the residual reconstruction error ε as the error in the k-space domain due to the error in motion estimates

$$\boldsymbol{\varepsilon} = \boldsymbol{E}\left(\boldsymbol{u}_{true}\right)\boldsymbol{x} - \boldsymbol{E}\left(\boldsymbol{u}\right)\boldsymbol{x} = \boldsymbol{E}\left(\boldsymbol{u} + \delta\boldsymbol{u}\right)\boldsymbol{x} - \boldsymbol{E}\left(\boldsymbol{u}\right)\boldsymbol{x}. \tag{13.11}$$

This expression can be expanded to

$$\boldsymbol{\varepsilon} = \left[\begin{array}{c} \mathbf{B}_1\mathbf{F}\mathbf{C}_1\left(\mathbf{M}_{u_1+\delta u_1}\boldsymbol{x} - \mathbf{M}_{u_1}\boldsymbol{x}\right) \\ \vdots \\ \mathbf{B}_{\mathrm{N}_m}\mathbf{F}\mathbf{C}_1\left(\mathbf{M}_{u_{\mathrm{N}_m}+\delta u_{\mathrm{N}_m}}\boldsymbol{x} - \mathbf{M}_{u_{\mathrm{N}_m}}\boldsymbol{x}\right) \\ \hline \vdots \\ \hline \mathbf{B}_1\mathbf{F}\mathbf{C}_{\mathrm{N_c}}\left(\mathbf{M}_{u_1+\delta u_1}\boldsymbol{x} - \mathbf{M}_{u_1}\boldsymbol{x}\right) \\ \vdots \\ \mathbf{B}_{\mathrm{N}_m}\mathbf{F}\mathbf{C}_{\mathrm{N_c}}\left(\mathbf{M}_{u_{\mathrm{N}_m}+\delta u_{\mathrm{N}_m}}\boldsymbol{x} - \mathbf{M}_{u_{\mathrm{N}_m}}\boldsymbol{x}\right) \end{array}\right]. \tag{13.12}$$

We can further expand this expression using the optical flow equation: Given an image $\boldsymbol{I}$ and a small displacement $\delta\boldsymbol{u}$, one can write $\boldsymbol{I}\left(\boldsymbol{u} + \delta\boldsymbol{u}\right) - \boldsymbol{I}\left(\boldsymbol{u}\right) \approx \nabla\boldsymbol{I}^T \cdot \delta\boldsymbol{u}$. Noting the motion-transformed images $\boldsymbol{x}_m = \mathbf{M}_{u_m}\boldsymbol{x}$ $(m = 1 \ldots N_m)$, we end up with

$$\boldsymbol{\varepsilon}\left(\delta\boldsymbol{u}\right) \approx \left[\begin{array}{c} \mathbf{B}_1\mathbf{F}\mathbf{C}_1(\nabla\boldsymbol{x}_1^T \cdot \delta\boldsymbol{u}_1) \\ \vdots \\ \mathbf{B}_{\mathrm{N}_m}\mathbf{F}\mathbf{C}_1(\nabla\boldsymbol{x}_{\mathrm{N}_m}^T \cdot \delta\boldsymbol{u}_{\mathrm{N}_m}) \\ \hline \vdots \\ \hline \mathbf{B}_1\mathbf{F}\mathbf{C}_{\mathrm{N_c}}(\nabla\boldsymbol{x}_1^T \cdot \delta\boldsymbol{u}_1) \\ \vdots \\ \mathbf{B}_{\mathrm{N}_m}\mathbf{F}\mathbf{C}_{\mathrm{N_c}}\nabla\boldsymbol{x}_{\mathrm{N}_m}^T \cdot \delta\boldsymbol{u}_{\mathrm{N}_m} \end{array}\right] = \left[\begin{array}{c} \mathbf{B}_1\mathbf{F}\mathbf{C}_1\boldsymbol{G}_1 \\ \vdots \\ \mathbf{B}_{\mathrm{N}_m}\mathbf{F}\mathbf{C}_1\boldsymbol{G}_{\mathrm{N}_m} \\ \hline \vdots \\ \hline \mathbf{B}_1\mathbf{F}\mathbf{C}_{\mathrm{N_c}}\boldsymbol{G}_1 \\ \vdots \\ \mathbf{B}_{\mathrm{N}_m}\mathbf{F}\mathbf{C}_{\mathrm{N_c}}\boldsymbol{G}_{\mathrm{N}_m} \end{array}\right]\delta\boldsymbol{u} = J_u\left(\boldsymbol{x}, \boldsymbol{u}\right)\delta\boldsymbol{u}. \tag{13.13}$$

We have introduced sparse matrices $\boldsymbol{G}_m$ $(m = 1 \ldots N_m)$, of size $N_x \times N_x N_m N_{dims}$, that select within $\delta\boldsymbol{u}$ the displacement field corresponding to the m$^{\text{th}}$ shot and apply the dot product with the image gradients $\nabla\boldsymbol{x}_m$.

Eq. (13.13) enables the residual reconstruction error to be predicted as a linear function of the error in the displacement fields $\delta\boldsymbol{u}$. The linear operator $J_u\left(\boldsymbol{x}, \boldsymbol{u}\right)$ is actually the Jacobian matrix of the error function that we used in the data fidelity term $f\left(\boldsymbol{x}, \boldsymbol{u}\right) = \boldsymbol{E}\left(\boldsymbol{u}\right)\boldsymbol{x} - \boldsymbol{s}$. In other words, $J_u\left(\boldsymbol{x}, \boldsymbol{u}\right) =$

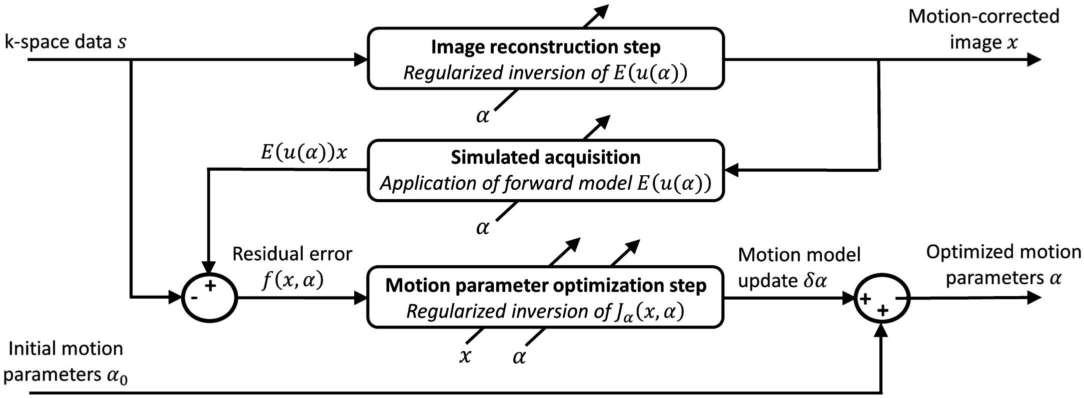

FIGURE 13.5

Graphical representation of the algorithm described for joint optimization of image and motion. From the motion-corrupted k-space data and an initial set of motion parameters (e.g., a rigid or a nonrigid motion model), a first image reconstruction step is performed that provides a first estimate of the image $\boldsymbol{x}$. Then the forward model is applied and the simulated k-space data are compared with the measured ones. The residual error in k-space domain is the input for the motion model optimization step that provides an update of the motion parameters minimizing the residual error. Then the motion model is updated and the process is repeated until a convergence criterion is satisfied.

$\nabla_u f(\boldsymbol{x}, \boldsymbol{u})$, i.e. it represents the partial derivatives of f with respect to the elements of the displacement fields.

If we further introduce a motion model $\boldsymbol{\alpha}$ that is a linear function of $\boldsymbol{u}$ (since that is the case for the two examples taken in this chapter), then we can write similarly

$$\varepsilon(\delta\boldsymbol{\alpha}) = J_\alpha(\boldsymbol{x}, \boldsymbol{\alpha})\,\delta\boldsymbol{\alpha}. \tag{13.14}$$

Detailed expressions for J_α will be given in Sections 13.2.3.3 and 13.2.3.4 in full-matrix formalism, which is not necessarily the preferred way for implementing these expressions, but which is extremely useful for understanding how the Hermitian transpose operators can be computed in a mathematically rigorous way.

13.2.3.2 *Alternating Gauss–Newton optimization*

Now we can formulate a general algorithm for solving Eq. (13.10). Considering an initial guess of motion defined by the motion model parameters $\boldsymbol{\alpha}$, we can proceed iteratively using the following scheme [69,71]: (i) Solve the optimization problem with respect to $\boldsymbol{x}$, considering $\boldsymbol{\alpha}$ fixed (using Eq. (13.8)); (ii) compute the residual reconstruction error $f(\boldsymbol{x}, \boldsymbol{\alpha}) = \boldsymbol{E}(\boldsymbol{u}(\boldsymbol{\alpha}))\boldsymbol{x} - \boldsymbol{s}$; and (iii) solve the optimization problem with respect to the motion model parameters $\boldsymbol{\alpha}$, considering $\boldsymbol{x}$ fixed. Then we repeat step (i) to (iii) until some stopping condition has been reached. A graphical representation of this optimization scheme is given in Fig. 13.5.

Step (iii) is the "Gauss–Newton" step, which involves the linearization around $\boldsymbol{u}(\boldsymbol{\alpha})$. Indeed, since the error function f is a nonlinear function of $\boldsymbol{\alpha}$, instead of directly searching for $\boldsymbol{\alpha}$ minimizing $f(\boldsymbol{x}, \boldsymbol{\alpha})$, the idea is to search for an optimal refinement $\delta\boldsymbol{\alpha}$ that minimizes $f(\boldsymbol{x}, \boldsymbol{\alpha} + \delta\boldsymbol{\alpha})$. The optimization with respect to the motion model is then formulated as

$$\begin{aligned} \hat{\delta\boldsymbol{\alpha}} &= \underset{\delta\boldsymbol{\alpha}}{\arg\min} \, \| f(\boldsymbol{x}, \boldsymbol{\alpha} + \delta\boldsymbol{\alpha}) \|_2^2 + \mu \, \| \boldsymbol{S} (\boldsymbol{\alpha} + \delta\boldsymbol{\alpha}) \|_2^2 \\ \hat{\delta\boldsymbol{\alpha}} &\approx \underset{\delta\boldsymbol{\alpha}}{\arg\min} \, \| f(\boldsymbol{x}, \boldsymbol{\alpha}) + J_\alpha (\boldsymbol{x}, \boldsymbol{\alpha}) \, \delta\boldsymbol{\alpha} \|_2^2 + \mu \, \| \boldsymbol{S} (\boldsymbol{\alpha} + \delta\boldsymbol{\alpha}) \|_2^2 . \end{aligned} \tag{13.15}$$

The least-squares solution is given by

$$\hat{\delta\boldsymbol{\alpha}} \approx \left(J_\alpha (\boldsymbol{x}, \boldsymbol{\alpha})^H J_\alpha (\boldsymbol{x}, \boldsymbol{\alpha}) + \mu \boldsymbol{S}^H \boldsymbol{S} \right)^{-1} \left(J_\alpha (\boldsymbol{x}, \boldsymbol{\alpha})^H (\boldsymbol{s} - \boldsymbol{E}(\boldsymbol{u}(\boldsymbol{\alpha})) x) - \mu \boldsymbol{S}^H \boldsymbol{S} \boldsymbol{\alpha} \right). \tag{13.16}$$

It should be noted that the motion parameters must be real numbers. However, the variables and operators on the right-hand side of Eq. (13.13) are complex, and therefore there is no guarantee that the solution $\hat{\delta\boldsymbol{\alpha}}$ will be purely real. Therefore, the motion model is updated using the formula $\boldsymbol{\alpha} := \boldsymbol{\alpha} + Re\left(\hat{\delta\boldsymbol{\alpha}}\right)$.

Since the linearization with the optical flow equation is only valid for small motion errors, in order to handle large displacements (especially when no initial guess of motion is available), a multiresolution strategy can be implemented. This means that the low-resolution k-space data (i.e., a central square) can be used first to reconstruct a low-resolution image (assuming motion is null), and then a low-resolution estimate of the motion parameters can be obtained. At the end of each resolution level, the motion model solution is scaled/interpolated to the next resolution level, providing an adequate initial guess for the higher resolution estimate.

13.2.3.3 Case of translational motion

In the case of translational motion, the motion parameters are defined as the translations occurring at each shot (in x and y dimensions, considering a 2D case for simplicity): $\boldsymbol{\alpha} = \left[\tau_1^{(x)} \ldots \tau_{N_m}^{(x)} \quad \ldots \quad \tau_1^{(y)} \ldots \tau_{N_m}^{(y)} \right]^T$. In matrix formalism, the relationship between the displacement fields and the motion model parameters is expressed as

$$\boldsymbol{u} = \left[u_1^{(x)} \ldots u_{N_m}^{(x)} \quad \ldots \quad u_1^{(y)} \ldots u_{N_m}^{(y)} \right]^T = \boldsymbol{P}\boldsymbol{\alpha}, \tag{13.17}$$

where $\boldsymbol{P}$ is a sparse matrix of size $N_x N_m N_{dims} \times N_m N_{dims}$ that duplicates the scalar translation value at each shot onto each pixel of the corresponding displacement field. Finally, the Jacobian matrix becomes

$$J_\alpha(\boldsymbol{x},\boldsymbol{\alpha}) = \begin{bmatrix} \mathbf{B}_1\mathbf{F}\mathbf{C}_1\boldsymbol{G}_1\boldsymbol{P} \\ \vdots \\ \underline{\mathbf{B}_{\mathrm{N}_m}\mathbf{F}\mathbf{C}_1\boldsymbol{G}_{\mathrm{N}_m}\boldsymbol{P}} \\ \vdots \\ \overline{\mathbf{B}_1\mathbf{F}\mathbf{C}_{\mathrm{N_c}}\boldsymbol{G}_1\boldsymbol{P}} \\ \vdots \\ \mathbf{B}_{\mathrm{N}_m}\mathbf{F}\mathbf{C}_{\mathrm{N_c}}\boldsymbol{G}_{\mathrm{N}_m}\boldsymbol{P} \end{bmatrix}. \tag{13.18}$$

This approach can be extended to include rotations as well; for more details, refer to Cordero-Grande et al. [21].

13.2.3.4 Case of a temporally constrained, nonrigid motion model

In the case of nonrigid motion, it has been proposed to use a separable motion model of the form $\boldsymbol{u}(r,t) = \sum_{k=1}^{N_k} v_k(r) w_k(t)$, where w_k $(k = 1 \ldots N_k)$ are known motion signals obtained from external sensors (e.g., respiratory sensors) or MR navigator data, and v_k are spatial coefficient maps controlling the local amplitude and direction of motion [68]. We also refer to Chapter 9 where the concept of partial separability has been discussed. We define $\boldsymbol{\alpha}$, to be the concatenation of all v_k maps, which have an x, y (and z) component like $\boldsymbol{u}$: $\boldsymbol{\alpha} = \left[v_1^{(x)} \ldots v_{N_k}^{(x)} \quad \ldots \quad v_1^{(y)} \ldots v_{N_k}^{(y)} \right]^T$. Hence, $\boldsymbol{\alpha}$ is a vector of $N_x N_k N_{dims}$ elements, and, if the number of motion signals is small compared to the number of motion states ($N_k \ll N_m$), and preferably if we have a certain degree of overdetermination (Nex > 1), this model can reduce the number of unknowns sufficiently to make the optimization tractable. In matrix formalism, the displacement fields can be written as $\boldsymbol{u} = \boldsymbol{Q}\boldsymbol{\alpha}$, with $\boldsymbol{Q}$ a sparse matrix of size $N_x N_m N_{dims} \times N_x N_k N_{dims}$, composed of diagonal block matrices, each diagonal having a constant value given by the motion signals w_k at the corresponding motion state m. Finally, the Jacobian matrix reads:

$$J_\alpha(\boldsymbol{x},\boldsymbol{\alpha}) = \begin{bmatrix} \mathbf{B}_1\mathbf{F}\mathbf{C}_1\boldsymbol{G}_1 \\ \vdots \\ \underline{\mathbf{B}_{\mathrm{N}_m}\mathbf{F}\mathbf{C}_1\boldsymbol{G}_{\mathrm{N}_m}} \\ \vdots \\ \overline{\mathbf{B}_1\mathbf{F}\mathbf{C}_{\mathrm{N_c}}\boldsymbol{G}_1} \\ \vdots \\ \mathbf{B}_{\mathrm{N}_m}\mathbf{F}\mathbf{C}_{\mathrm{N_c}}\boldsymbol{G}_{\mathrm{N}_m} \end{bmatrix} \boldsymbol{Q}. \tag{13.19}$$

13.3 Methods

We will now focus on how the three families of motion correction techniques that we have described can be implemented in practice. The k-space phase correction technique described in Section 13.2.1

(or its extension to rigid motion) has been widely used, despite its limitations, because it is the simplest and most computationally efficient. As for the general motion-corrected reconstruction described in Section 13.2.2, efforts need to be made in order to estimate the motion parameters corresponding to each k-space sample, shot or motion state. Even for the joint reconstruction of image and motion, in the nonrigid case, some prior information about motion is necessary, in the form of motion signals. Therefore, we will review some of the strategies that can be used to provide motion estimates. Generally speaking, motion information can be obtained in the form of time-varying scalar quantities (e.g., translation parameters given by a navigator measurement) or time-varying vector fields of one, two or three spatial dimensions (e.g., from the registration of image navigators). Some of these motion signals may provide a direct motion measurement of the organ of interest, while others may only provide a surrogate measurement (e.g., an external sensor measurement or motion from a remote body region). In the latter case, they need to be related to the organ motion through a motion model. The choice of the k-space sampling scheme will also be discussed since it is particularly important for the motion-estimation step, in particular, for the joint reconstruction of image and motion. Finally, we will also discuss the connection between motion estimation/correction and accelerated imaging, i.e., how it can help or improve the solving of highly undersampled MRI data for applications in dynamic MRI with high temporal resolution or even real-time MRI.

13.3.1 Strategies for motion sensing

13.3.1.1 External sensor measurements

External sensors are routinely used in clinical cardiac MRI. The electrocardiogram (ECG) is most commonly used (an alternative is the peripheral pulse oximetry) as a cardiac motion signal. From this signal, a cardiac phase can be derived that is a temporal position in the cardiac cycle, assumed to be periodic, up to a stretching of the systole and diastole periods. The ECG is generally used only to synchronize the acquisition and/or the reconstruction with the heart beating, but the cardiac-phase signal can also be used as a surrogate signal for cardiac motion in the vicinity of a given cardiac phase [88].

Respiratory belts or bellows are also provided by the MRI manufacturers. The associated signals are approximately proportional to the amplitude of chest surface or abdominal surface deformation, according to their placement on the patient. Either the amplitude or the phase of the respiratory signal can be used as an input signal for a predictive-motion model. Accelerometer-based sensors (actually used as inclinometers) have also been proposed for obtaining respiratory signals [14].

For tracking head motion in MRI, optical systems have also been extensively developed [31,58], but they have been mostly used with prospective correction rather than retrospective correction. Such systems aim to track the displacement of reflective markers placed on the head with infrared cameras. Multiple cameras can be used with a stereoscopic reconstruction. More recently, single-camera systems have been proposed, with the camera located inside the bore, close to the patient. They use particular markers such as encoded checkerboards or moiré patterns in order to resolve motion. These systems aim to provide rigid head-motion parameters with six degrees of freedom (three translations and three rotations).

13.3.1.2 Extracting motion from MR data

Motion can be extracted from the MR data in multiple ways. In MRI, an image of interest is acquired by repeating many pulse sequence blocks. Separate pulse-sequence blocks can be added and interleaved with the imaging pulse-sequence blocks. In that case, the added blocks are dedicated to motion estimation. This strategy will be termed "separate navigation" in the remainder of the section. Another strategy is to use the data from the imaging pulse-sequence blocks themselves to extract motion information. This strategy will be termed self-navigation. For both strategies, the extracted motion exformation can be of low spatial dimension: It can be dimensionless (a point-like particle with no spatial dimension), i.e., represented by a time-varying scalar term such as the k-space center, or it can be 1D (one spatial dimension), i.e., a vector of time-varying parameters or a time-varying line through the k-space or through the image. It can also be of higher spatial dimension (2D or 3D), and in that case it makes it possible for a 2D or 3D image to be acquired and post-processed in order to estimate more complex motion information, such as motion in multiple directions or local deformations. Finally, we will also describe alternative ways using built-in or additional MRI hardware or those based on the analysis of the acquired noise. An illustration of these methods is given in Fig. 13.6 for cardiovascular applications.

Separate navigation signals

Dedicated MR sequence blocks interleaved in the imaging sequence can provide motion tracking data. Such navigator blocks are available on clinical scanners for cardiac or abdominal applications. Typically, they consist of exciting a pencil-beam region placed on the diaphragm and aim to track the displacement of the interface between lung and liver (see Fig. 13.6). Navigators have also been extensively developed for head applications to track rigid motion [31,58], including orbital, spherical, and cloverleaf navigators, in the effort to identify the six motion parameters with the minimal amount of extra k-space data.

Navigator data can also be obtained by a small modification of the imaging sequence block, which results in a minor increase of the repetition time. The so-called DC navigator (or FID navigator) consists of opening the MR data acquisition window between the slice selection and the phase-encoding gradient, which corresponds to sampling the k-space center point ($k_x = k_y = k_z = 0$) [12]. This sample corresponds to the sum of all pixel intensities in the image. With a receiving surface coil, it is typically modulated by anatomical changes such as bulk or breathing motion. Another strategy consists of sampling MR data during the prewinder and/or rewinder gradients. This amounts to sampling the k-space with a butterfly-shaped trajectory, which can be tuned to retrieve directional motion-information (bulk motion in x, y, z) [15].

All these navigator data come as multicoil k-space data so, even if they provide a global information from the whole image, it is inherently localized to a body region close to the receiving coil, due to the coil sensitivity weighting. Therefore, the analysis of such data often comes with either a coil selection criterion or more generally, a principal component analysis combined with a frequency analysis, in order to extract signals of cardiac or respiratory origin.

Separate image navigation for 2D/3D motion estimation

Navigators providing low spatial-resolution images have also been proposed for tracking head motion, e.g., using fast spiral scans in the plane of interest [8] or in three orthogonal planes [89]. Similarly, for cardiovascular or abdominal applications, 2D or 3D image navigators have been proposed [1,39,47,57]. A specificity of cardiac imaging is that, often, image acquisition is prospectively synchronized to the

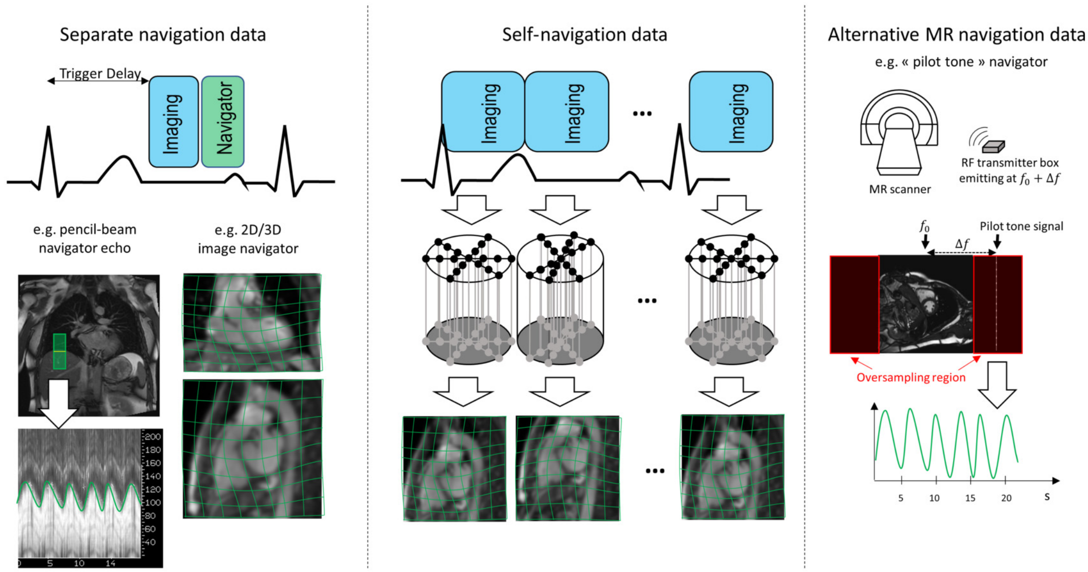

FIGURE 13.6

Illustration of a few methods available for extracting motion information from MR data for cardiovascular applications. On the left, separate MR navigation data are obtained from separate pulse-sequence blocks, dedicated to motion monitoring. They can provide a 1D image (e.g., a pencil beam navigator placed on the diaphragm), or a low-resolution 2D or 3D image that can be used to track more complex motion. In the middle, self-navigated imaging typically uses radial/spiral sampling (or radial/spiral phase encoding in 3D Cartesian sequences). They can also provide a rough motion signal (using the k-space center, or center line, which is sampled frequently), or a low-resolution 2D or 3D image reconstructed from undersampled k-space data acquired in similar motion states (e.g., consecutive spokes or respiratory-binned spokes). On the right, the pilot tone navigator is an example alternative MR method. An RF transmitter emits at a frequency intended to be outside the imaging bandwidth, corresponding to the field of view of interest, but still within the measured MRI data (oversampling region). The pilot tone signal is modulated by patient motion, which allows a motion signal to be extracted.

heart rate so that the imaging data are acquired in the end-diastolic rest period (see Fig. 13.6). This means other periods of the cardiac cycle can be filled with motion tracking-sequence blocks to get heart-motion information in all three directions, as well as nonrigid components of motion.

Self-navigation signals

The imaging sequence can be designed to acquire the central k-space data of the imaging experiment repeatedly, at a sufficiently fast rate, in order to get motion tracking data. This feature is intrinsic to radial or spiral sequences since the center k-space point is acquired at every repetition. With Cartesian trajectories, in particular (but also with radial trajectories), the center k-space line can be interleaved in the sequence [51,80,82], providing, after using a Fourier transform, a projection of the image onto the frequency encoding axis, from which cardiac/respiration signals can be derived. Trajectories made of rotated Cartesian tiles, as used by the PROPELLER technique [73], allow a full disk to be acquired at the center of k-space, i.e., providing a low-resolution image at each shot.

Furthermore, in some cases, the imaging data themselves allow more complex 2D or 3D motion information to be extracted, based on smart sampling strategies in the k-space. The radial trajectory has been very popular for motion-corrected reconstruction. Indeed, it is generally designed with a specific angular step between successive radii based on the golden ratio (111.246 degrees) [90]. This scheme allows a flexible choice for the reconstruction of dynamic data since k-space data can be grouped in multiple ways, each way providing a relatively uniform k-space sampling: either small blocks of consecutive radii can be formed, providing low-spatial, high-temporal resolution images; or large blocks can be formed, providing high-spatial, low-temporal resolution images (see Fig. 13.6). The high spatial resolution images are typically corrupted by motion due to the large temporal footprint. The fast, low spatial resolution images can therefore be used to estimate motion fields, by image registration, and this enables a motion-corrected reconstruction of the high spatial-resolution data. In 3D applications, similar properties are obtained by using a Cartesian sampling with a pseudo-radial or pseudo-spiral sampling of the k_y-k_z plane. Such sampling strategies have been very popular for respiratory and/or cardiac motion correction [24,36,43,72,75,79,83]. Often, in such applications, a fixed number of motion states is determined based on various types of motion signals (ECG, respiration sensors, self-navigation data). The k-space data can then be split into different sets, each corresponding to one motion state. For instance, a respiration signal may allow a number N_m of motion states to be defined by binning the data according to the amplitude of the motion signal. Then an image reconstruction technique suited for (possibly highly) undersampled data is employed to form images, which can be of moderate quality but sufficient to estimate motion by image registration.

Alternative MR navigation data

Small dedicated radiofrequency receiver coils, also termed pick-up coils or NMR field probes, can also be placed on the patient to provide real-time positional information [31]. While initial systems required additional gradients to measure the NMR probe position, simultaneous imaging and field-probe measurements have been demonstrated with ^{19}F probes and with carefully designed gradient waveforms [3,33]. It has also been proposed to probe the magnetic field gradients of the imaging sequence with a three-axis Hall magnetometer, in order to derive its location in the scanner, and therefore to estimate patient motion [86].

Another strategy consists of placing a small radiofrequency transmitter, called pilot tone, emitting at a frequency outside the imaging bandwidth but still within the receiving coil bandwidth [85]. Due to the interaction between the pilot-tone transmitter and the patient, the pilot-tone signals measured by each coil are modulated by patient motion (see Fig. 13.6).

Motion also modulates the variance of the thermal noise associated with the k-space data measurements, and these changes can be measured [2].

13.3.2 Image registration

Image registration is the process by which motion parameters can be estimated from images in two or more motion states. There is a rich literature in digital image processing applied to medical image and many freely available software tools [35,48,62,63]. Since we aim to form the forward model described in Section 13.2.2.2, it is the image in the reference motion state that should be chosen as the floating image, i.e., which should be registered onto the image at each other motion states (i.e., the target images). When choosing a particular registration technique, several aspects should be considered. Firstly, we should consider how motion is parameterized, e.g., with a rigid/affine transformation, with a dense displacement field, or with free local displacements at some control points with spline interpolation. Then, we need to choose a similarity metric to quantify to spatial alignment of the images. The sum of squared differences (i.e., the L_2 norm of the difference image) is the natural choice (leading to a computationally efficient optimization) when there are no (or minor) intensity changes between the images. Other metrics, such as the correlation coefficient or the normalized mutual information, are popular choices otherwise.

13.3.3 Motion models

When the motion parameters of interest are available (e.g., six parameters of rigid body motion or deformable motion field), they can be used directly as inputs of a motion-corrected reconstruction. Otherwise, surrogate motion data can be used in order to estimate the motion parameters or the deformable motion fields. We have seen indeed many ways of collecting motion signals, but often these signals do not provide a quantitative measure of motion in mm units (e.g., respiratory belt, DC navigator, pilot tone, etc.) and/or they provide a measure at the body surface or in another organ such as the diaphragm. So they do not provide a direct measure of motion for the organ of interest. However, it may be assumed that they are temporally correlated with the motion of the organ of interest. Let us consider that we seek to estimate N_p motion parameters at a given time t, i.e., a vector $p(t)$, which can be, e.g., rigid motion parameters ($N_p = 6$) or a displacement vector field ($N_p = N_x N_{dims}$, i.e., the displacement in x, y, z for each voxel). A linear model can be introduced to relate $p_i(t)$ $(i = 1 \ldots N_p)$ to the measured physiologic motion signals $\varphi_k(t)$ $(k = 1 \ldots N_k)$ [59,65]

$$p_i\,(t) = \sum_{k=1}^{N_k} a_{i,k} \varphi_k(t). \tag{13.20}$$

The $a_{i,k}$ are some coefficients to be determined. A calibration scan can be implemented to acquire the motion signals $\varphi^{calib}\,(t)$ simultaneously with fast (possibly low-resolution) imaging data. Then the fast imaging data are registered with a rigid or nonrigid registration technique according to the desired type of motion model, thus providing $p^{calib}\,(t)$. The $a_{i,k}$ are estimated by linear regression. Finally, the motion model can be used during the sequence of interest to predict the motion parameters from the new motion signals, using Eq. (13.20). This linear model has been the most widely used in MRI, but more general models can be designed [60].

13.3.4 Optimal k-space sampling for motion correction

The joint reconstruction of image and translational (or rigid) motion has been shown to be feasible without any additional motion measurement. However, the k-space sampling scheme has an impact on the accuracy of motion estimation and thereby the performance of the reconstruction. A uniformly distributed, and even incoherent, sampling of k-space should be preferred to a linear sampling since it results in a much more accurate reconstruction of the motion parameters [19]. A similar observation has already been made in Section 13.3.1.2 in the case of a radial sampling strategy, with the golden angular step ensuring a relatively uniform, but also incoherent, sampling of k-space in each motion state. In that case it is well understood that such a sampling scheme is a necessary condition for the reconstruction of images from each motion state using a compressed sensing or low-rank technique. The quality of those images indeed impacts the accuracy of the subsequent motion estimation (i.e., the registration of images from each motion state).

13.3.5 Motion correction to improve dynamic MRI

When imaging a moving patient, reconstructing a static, motion-corrected image is one possible approach to the problem, but another approach is to reconstruct a dynamic imaging dataset by thinking of the acquisition domain as a k-t space, rather than a k-space. When is it better to use one approach or the other? Is there any connection between them, and can we combine them?

The answer to the first question may depend on the target application. On the one hand, if we need some quantitative information from the static image only, motion correction may be more relevant since the combination of all motion-corrupted data will lead to a higher SNR. On the other hand, if we seek to quantify the dynamic information, as in cardiac cine imaging (contractile function assessment), and if many parameters are required to describe motion, one may think it is more efficient to encode the whole k-t space (and possibly several temporal dimensions for cardiac and respiratory motion as in XD-GRASP [27]) than a static k-space and the motion parameters. However, it was observed in [74] that, when the displacement fields of the moving structures were relatively smooth in time, the temporal bandwidth associated with the motion parameters was lower than that associated with the pixel intensities. Indeed, pixels at the edge of a moving organ underwent a sharp intensity transition that resulted in high temporal frequencies in the k-t space. Thus, it was shown in Prieto et al. [74] that a motion model could be fit to undersampled k-t space data, which allowed the full cardiac cycle to be modeled. The motion model was then applied to fill in the missing k-space data. This approach allowed high undersampling factors.

It was further proposed to integrate motion estimates into a compressed sensing reconstruction of undersampled data in the k-t space, namely k-t FOCUSS, for cardiac cine imaging [45,46]. The authors noted that the concept of motion estimation and compensation was present in video compression, such as MPEG, and could improve the encoding of the dynamic information. The method in Jung et al. [46] generalized the classic k-t FOCUSS by introducing a prediction of the various imaging frames, using motion estimates between certain key frames of the series (obtained by a block matching technique). Therefore, only the residual dynamic imaging data needed to be encoded in the k-t space. As a result, the encoded data in k-t space were sparser, and higher acceleration factors could be achieved. This concept has been used with different combinations of algorithms for the image reconstruction step and for the motion estimation step [4,83].

A similar concept was proposed in real-time radial MRI [52], with applications in cardiac and speech imaging. In that work, images were first reconstructed from highly undersampled radial data using a specific technique (so-called “nonlinear inversion” that reconstructs the image and coil sensitivities jointly). Then data from five consecutive frames were combined: Motion between these frames was estimated by an optic flow registration technique; then motion was integrated into the image reconstruction step. Motion correction was thus shown to improve the temporal fidelity of the real-time reconstruction.

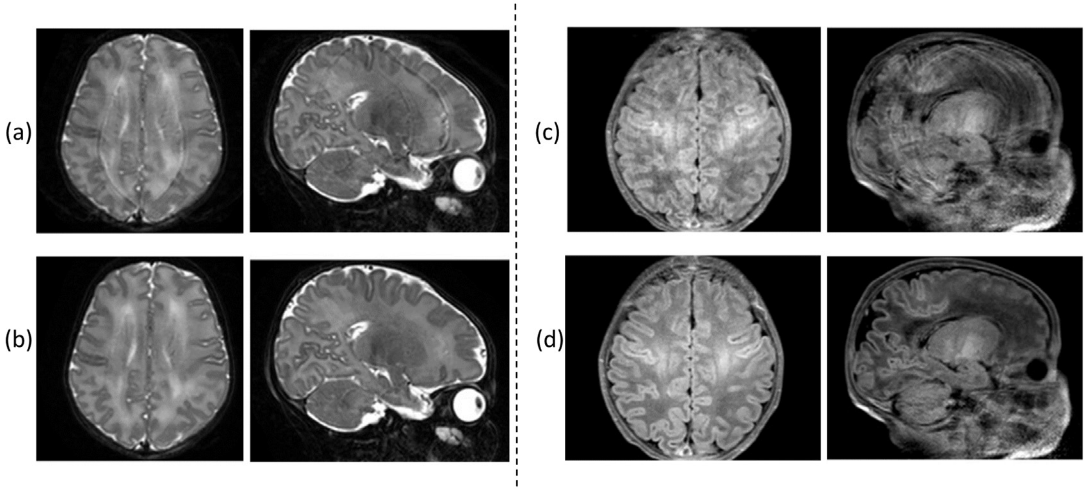

FIGURE 13.7

Example neonatal brain images without (respectively with) motion correction by joint reconstruction of image and rigid motion parameters: T_2-weighted (**a** and **b** respectively) and T_1-weighted (**c** and **d** respectively) multishot multislice FSE sequences. Extracted from Cordero-Grande et al. [20].

13.4 Clinical application examples

We will now illustrate the described theory and method with some clinical applications. Rather than providing an exhaustive list, we aim to show some representative examples in various organs and using various types of motion inputs: no motion prior at all, motion estimates from low-resolution or self-navigation images, or motion prior from external sensors or navigators.

13.4.1 Brain

PROPELLER, thanks to its self-navigated k-space trajectory, is one the earliest motion-corrected reconstruction techniques and is currently available on clinical scanners (also called BLADE). It was shown to improve the quality of clinical images in FLAIR, T_2-, T_1- and/or contrast-enhanced T_1-weighted imaging compared to uncorrected, conventional scans. Various populations were studied including adults/elderly patients [66,91] and children [87]. Applications to high-resolution diffusion imaging have also been demonstrated in acute cerebral infarction [29] and diffusion tensor imaging [18].

The joint reconstruction of image and rigid motion proposed in Cordero-Grande et al. [20] has the great advantage of being applicable to conventional sequences so it does not lengthen the imaging protocol. It was successfully applied to a very large database of neonatal brain scans (>1800 volumes from >500 babies, gestational age between 32 and 45 weeks). The babies were scanned during natural sleep, and the 3D motion correction framework was combined with outlier rejection in order to discard shots with extreme artifacts. This database included T_1- and T_2-weighted multishot, multislice, fast spin-echo sequences (see examples in Fig. 13.7).

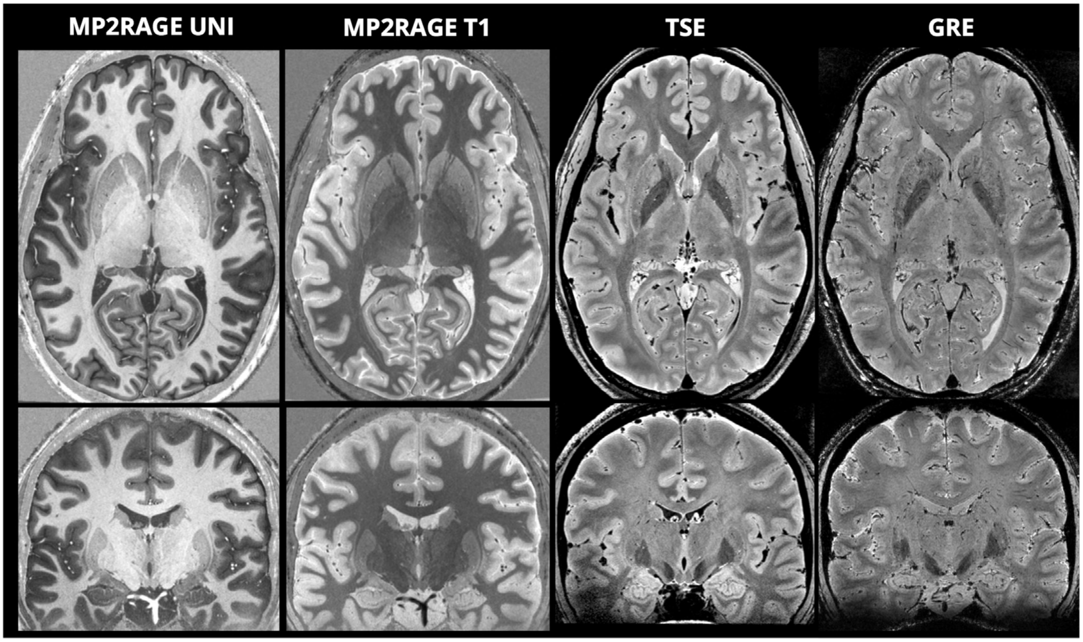

FIGURE 13.8

Ultra-high resolution brain MRI at 7T with motion-corrected reconstruction: T1-weighted image (MP2RAGE UNI), quantitative T_1 map (MP2RAGE T_1), Turbo-Spin Echo (TSE), and Gradient-Recalled Echo (GRE). Extracted from Federau and Gallichan [26].

In conventional multishot diffusion MRI sequences, an image navigator is generally acquired in order to obtain a shot-dependent phase map, required to combine the multiple shots. This low resolution navigator image has been further used for rigid motion estimation and allowed motion-corrected reconstruction of the multishot diffusion images [32,81].

Motion-corrected reconstruction also allowed ultra-high resolution human brain images at 7T with T_1-weighted, T_2-weighted and T_2^*-weighted sequences at 350 to 380 μm nominal isotropic resolution [26] (see Fig. 13.8).

13.4.2 Cardiovascular

Motion-corrected reconstruction has been an application of choice for high-resolution, 3D whole-heart coronary MR angiography [10,13,22,25,30,42,72,75,79]. Indeed, visualizing the coronaries typically requires lengthy 3D cardiac-gated scans and respiratory correction to achieve, ideally, submillimeter spatial resolution. These studies typically use radial/spiral sampling, or 3D Cartesian sampling with a radial/spiral sampling order, respiratory binning to define the motion states, and image registration to estimate motion. Throughout these studies, nonrigid correction was shown to improve vessel sharpness compared to rigid correction or conventional respiratory-gated solutions (see examples in Fig. 13.9).

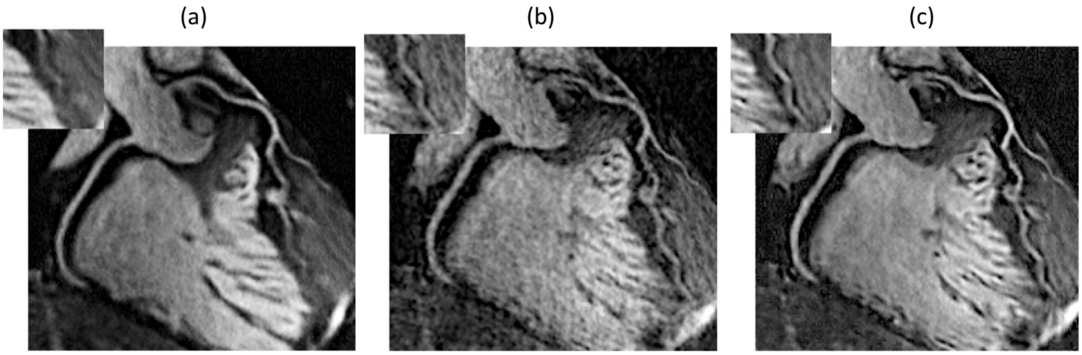

FIGURE 13.9

Example reformatted view of a 3D whole heart coronary MR angiography, with 0.9-mm isotropic resolution, obtained from free-breathing, ECG-gated data (3D balanced SSFP sequence with pseudo-spiral sampling). Various motion-corrected reconstruction techniques are shown: translational motion correction with a patch-based image regularization technique (PROST) (**a**); nonrigid motion correction (**b**); nonrigid correction with patch-based image regularization (**c**). Extracted from [13].

Motion correction also allows highly efficient acquisition, i.e., using 100% of the acquired data, as opposed to conventional gating approaches that reject data from undesired motion states [34]. This also makes the scan time predictable (i.e., not depending on how regular the breathing motion is).

Cardiac function assessment by cine imaging has also been a popular application. It is generally achieved clinically by a multi-slice coverage of the ventricles with a cardiac-gated balanced SSFP sequence. Nonrigid motion-corrected reconstruction has enabled free-breathing protocols using either radial sampling schemes and self-navigation (registration of extracted low resolution images) [36,83], or motion signals as a prior for a joint reconstruction of image and a nonrigid motion model (GRICS technique), including external sensors [88] and center k-space line navigators [70]. Clinical studies in adult patients have shown that such free-breathing, motion-corrected images are in good agreement with breath-held protocols in terms of volumetric parameters, i.e., end-diastolic and end-systolic volumes, stroke volume, and ejection fraction [23,88]. Free-breathing cardiac MRI assessment (function and tissue characterization) was also shown in Duchenne muscular dystrophy patients (from children to young adults), a population that has severely impaired respiratory function and therefore can only perform very limited breath-holds [11,67] (see example images in Fig. 13.10). Application to 4D flow MRI was also shown in pediatric congenital heart disease patients [16].

Extensions of the nonrigid motion correction framework to multicontrast datasets have also been proposed for free-breathing dynamic contrast enhanced (DCE) cardiac imaging [28] and for T_1 and T_2 mapping [61,69].

Multimodal cardiac imaging by combined PET/MRI systems is another appealing application. Indeed, the high spatial resolution of MRI and its soft tissue contrast makes it a better modality for motion tracking, which allows the simultaneously acquired PET images to be corrected using similar motion-corrected reconstruction techniques [49]. Nonrigid correction made it possible for coronary MR

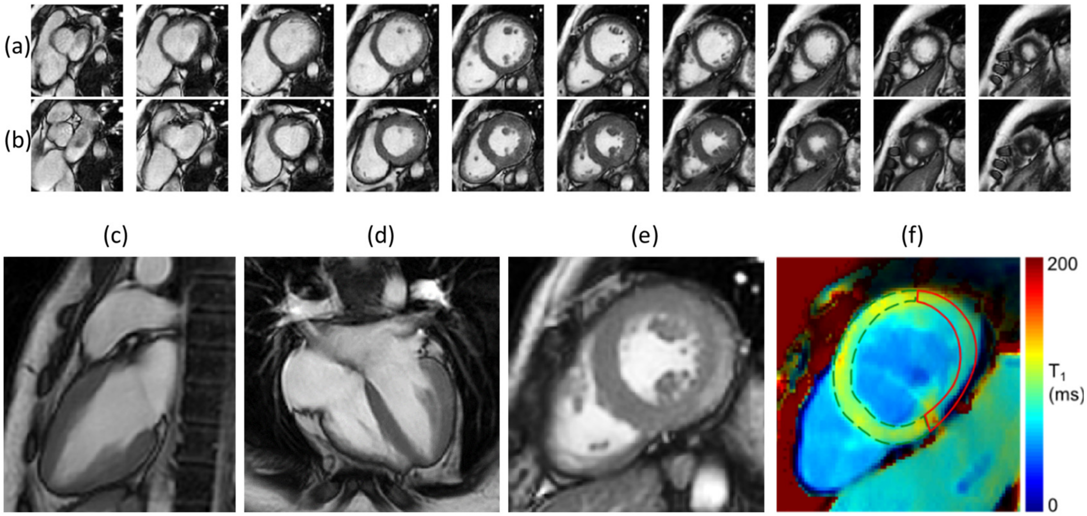

FIGURE 13.10

Example cardiac MR images from a sixteen-year-old Duchenne muscular dystrophy patient, using free-breathing and nonrigid motion correction (GRICS technique): 2D multislice cine (balanced SSFP sequence) in diastole (**a**) and systole (**b**); post-contrast 2D cine imaging in vertical, horizontal, and short axis views (**c**, **d**, **e** respectively); short axis T_1 map from a saturation recovery sequence (**f**). Data from clinical studies in Bonnemains et al. [11] and Odille et al. [67].

angiography to be obtained together with myocardial viability assessment by ^{18}F-FDG PET in patients with chronic total occlusion [64].

13.4.3 Body imaging (other than brain and heart)

Other body regions have also benefited from motion-corrected reconstruction, in particular, when high resolution 3D imaging and/or DCE imaging is necessary. Applications to 3D liver DCE imaging [53,43,44] have been shown, with the respiratory motion correction allowing quantitative perfusion maps to be calculated using Tofts or extended Tofts models (see example in Fig. 13.11). Nonrigid correction in 3D liver scans was also shown to improve the image quality in healthy subjects [24] and pediatric patients [17]. Application to free-breathing UTE imaging of the lung was shown in adults and pediatric patients, and nonrigid correction enabled better depiction of the lesions compared to conventional motion management techniques [93,94].

13.5 Current challenges and future directions

Despite significant advances over the last years, there are still several remaining challenges and unanswered questions in motion-corrected MRI reconstruction. One of the main challenges remains the

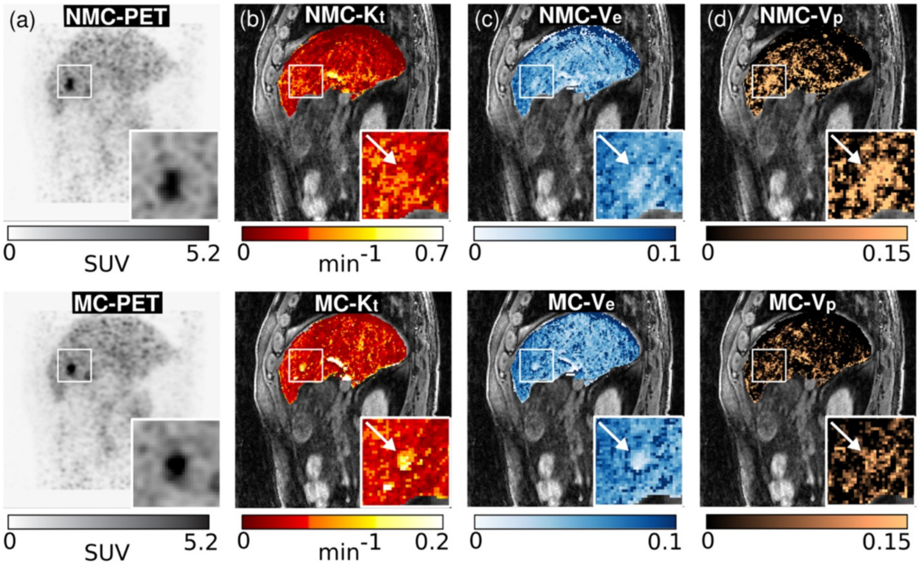

FIGURE 13.11

Example simultaneous acquisition of PET and DCE-MRI of the liver in a patient (lesion highlighted with the white arrow), injected with gadoxetate disodium. The authors used a 3D fat-suppressed T_1-weighted gradient-echo sequence with pseudo-radial sampling. PET images and parametric permeability maps (K_t (**b**), V_e (**c**) and V_p (**d**) fits from an extended Tofts model) are shown without motion correction (NMC, top row) and with nonrigid motion correction (MC, bottom row). Extracted from Ippoliti et al. [44].

increased computational complexity of motion-corrected reconstruction. Even in the case of known motion, like for other iterative MR reconstruction techniques, the computation time is mainly determined by the number of calls to the $\boldsymbol{E}^H\boldsymbol{E}$ matrix, and more precisely by the number of Fourier transformations. Compared to uncorrected reconstruction, this number is roughly multiplied by the number of motion states. In the case of a joint motion estimation, there is another added computational burden due to the motion estimation step.

Deep-learning methods have the potential to speed up different aspects of the motion-corrected reconstruction, and initial work has been done in that direction. A convolutional neural network (CNN) can be designed and trained to speedup the motion estimation step. Such CNNs have been proposed to estimate motion between pairs of patches from two images to be registered [77] or to estimate motion from a motion-corrupted patch [37]. Methods have also been proposed to model the reconstruction step as well [50]. An end-to-end deep-learning framework, consisting of a diffeomorphic registration network and a motion-informed model-based deep-learning reconstruction network, has been also recently proposed for nonrigid motion-corrected reconstruction of nine-fold undersampled, free-breathing, whole-heart coronary MR angiography [77]. Alternatively, a generative adversarial network (GAN) has been proposed to model the motion-corrected reconstruction problem without explicitly modeling motion [84]. The idea underlying GAN is that it includes two neural networks: a generator, here designed to produce motion-corrected images from corrupted input images, and a discriminator,

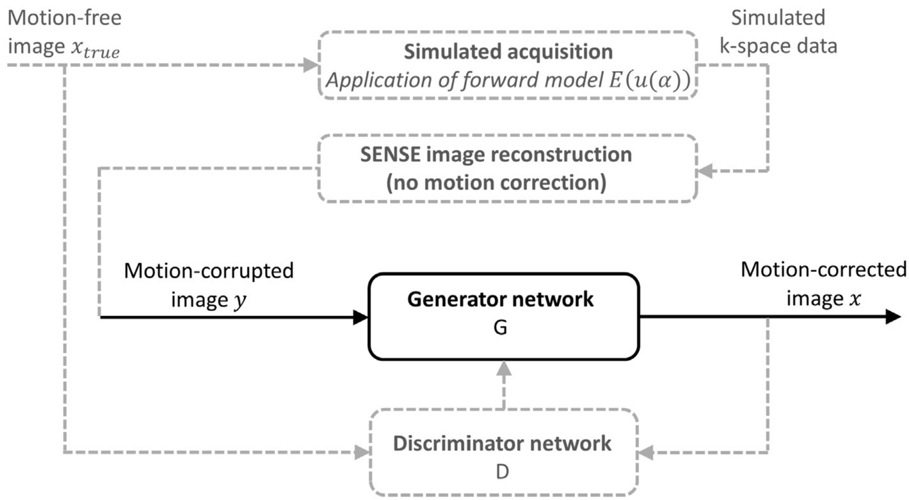

FIGURE 13.12

Example strategy for modeling the whole motion-corrected reconstruction problem with deep-learning, using a generative adversarial network (GAN), as proposed in Usman et al. [84]. The schema shows the motion-corrected image reconstruction, as modeled by a generator network, in solid black lines. The dashed gray lines indicate the elements needed for training the generator. This involves: a database of reference, motion-free images; a forward model of the motion-corrupted scan, including random motion parameters, together with a conventional (uncorrected) image reconstruction, that are used to synthetize corrupted images; these synthetic, corrupted images are the inputs of the generator, which is designed to produce motion-corrected images; finally, a discriminator network attempts to recognize which images are "true" (i.e., similar to those from the reference database) or "fake" (i.e., still motion-corrupted).

designed to distinguish between the generated images and some reference images without motion artifacts. In Usman et al. [84], the following strategy was proposed to train the GAN: a database of brain images, free of motion artifacts, was selected; motion-corrupted images were synthesized using a forward acquisition model similar to that described in Section 13.2.2.2, using a wide range of motion parameters (rigid motion); the synthetized images were used as input of the generator; and both the synthetized and reference (motion-free) images were used as inputs as the discriminator. See Fig. 13.12.

The joint estimation of image and motion can be solved very accurately in the case of rigid motion, possibly with no sequence modification (though optimizing the k-space trajectory is preferable). In the nonrigid case, authors have all used motion models, either implicitly (e.g., respiratory binning) or explicitly (e.g., as in GRICS). Such models only provide an approximate of the true motion, which thereby limits the achievable accuracy of the reconstruction, however they render the optimization tractable. A more general parameterization of motion, using a low-rank model for the full spatiotemporal displacement fields, was recently proposed in Huttinga et al. [41,40] and showed very promising results. How accurate the motion model can be remains an open question since a ground truth for the motion is difficult to obtain.

As mentioned in the introduction of this chapter, the impact of motion on MR physics is complex. We have assumed that the spatial encoding inconsistencies were the main effect, but this may not be the case in all applications. Patient motion also induces changes in B_0 field maps, especially at high field levels. This can be a concern with T_2^*-weighted imaging at 7T [55]. It is also a problem at 3T with balanced SSFP sequences in cardiac or abdominal applications since off-resonance artifacts (so-called banding artifacts) appear as a local distortion of the complex image and do not strictly follow the displacement of the neighboring organs, making motion estimation challenging. We have also implicitly assumed that the receiving coils were static and that the associated B_1^- field did not change with motion. The latter point means coil-loading changes were disregarded, however changes in sensitivity "seen" by moving tissues were intrinsically modeled in $\boldsymbol{E}$. Through-plane motion is another concern, but either 3D-encoded or stacks of 2D sequences can be used in order to model the actual 3D displacement. Besides artifacts in contrast-weighted images, motion can also lead to a bias in the quantitative MRI parameters. More generally, spin-history effects should always be kept in mind. Sequences with motion-compensated gradients can be used to mitigate those effects. The balanced SSFP sequence actually has a good motion-robustness in that regard, and it provides a relatively stable signal in presence of motion (due to its balanced gradients), which is part of the reasons why it is widely used in clinical cardiac protocols. Conversely, some sequences like fast spin-echo or diffusion-weighted imaging are known to be particularly sensitive to motion since it can create signal drops due to intravoxel dephasing. To tackle all these issues, should our future efforts be focused on combining prospective and retrospective motion correction techniques? Or should we rather aim at measuring all these motion-induced effects and modeling them into the reconstruction?

13.6 Summary

General motion (rigid or nonrigid) can be integrated in the MRI reconstruction in a mathematically rigorous way by embedding linear motion operators in the forward model. Various strategies can be used to estimate motion and thereby implement motion-corrected reconstruction. In the case of rigid motion, a blind reconstruction of the motion-free image and motion parameters is feasible in brain imaging and particularly effective when the k-space sampling scheme is uniform and incoherent. In the case of nonrigid motion, prior information is still needed in order to determine the number of motion states and to estimate motion between them. This prior can be obtained from separate image navigators or from self-navigation strategies (particularly with radial or 3D pseudo-radial/spiral trajectories). Alternatively, the joint reconstruction of a motion-free image and a time-constrained motion model is possible, using prior from 1D motion signals (e.g., external sensors or navigators). Applications in cardiac and body imaging include free-breathing imaging of lengthy 3D or dynamic sequences.

13.7 Practical tutorial

A demo code (Matlab®, The MathWorks, Inc., Natick, MA) illustrating the various types of motion-corrected reconstruction techniques, in synthetic data, is provided with this chapter, including: phase correction of translational motion in k-space; motion-corrected reconstruction using motion operators (rigid and nonrigid case); joint reconstruction of image and motion (translational motion and temporally constrained nonrigid motion). To run the tutorial, open the file "motion_corrected_reconstruction_demo.m", edit the simulation options in the first lines (number of shots, number of excitations, type of motion, type of k-space sampling), and run.

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